SHORT SUMS OF FACTORIAL DIVISORS AND ERDŐS PROBLEM 18(B)

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Abstract. For a practical integer N, let $h(N)$ be the largest, over $1 \le R \le N$, of the minimum number of distinct positive divisors of N whose sum is R. We prove that, for every $\delta > 0$, one has $h(n!) < n^{\delta}$ for all sufficiently large n, giving an affirmative answer to Erdős problem 18(b). More generally, for each $\eta > 0$, every positive integer R is a sum of $O_{\eta}((1 + \log\log(R+3))^{K_{\eta}})$ distinct divisors of a factorial whose index is $O_{\eta}((\log(R+3))^{1+\eta})$. The construction combines binary blocks of actual factorial divisors, multiplicative mixing modulo powers of two, and a positive quotient recursion. A factorial reserve removes repeated summands. A final factorial radix argument supplies the bound uniformly for every target up to $n!$. We give a proof using Bourgain's general-modulus exponential-sum theorem and describe a companion, single-file Lean proof of the exact formal conjecture. The Lean development proves its needed dyadic mixing estimate internally and has passed local type, axiom, and dependency audits.

1. Introduction

A positive integer N is *practical* if every integer in $[1, N]$ is a sum of distinct positive divisors of N. For such N, put

$$h(N) = \max_{1 \le R \le N} \min\left\{ |D| : D \subseteq \{d \in \mathbb{N} : d \mid N\}, \quad \sum_{d \in D} d = R \right\}. \tag{1.1}$$

Here all divisors and summands are positive. The empty sum is allowed when the target is zero. Factorials are practical; we recall the elementary induction in Section 6.

Erdős asked whether $h(n!)$ grows more slowly than every positive power of n; see Erdős and Graham [4] and the precise formal statement in [5, 3]. We prove this bound.

Theorem 1.1. *For every real $\delta > 0$ there is an integer $n_0(\delta)$ such that*

$$h(n!) < n^{\delta} \qquad (n \ge n_0(\delta)).$$

Equivalently, $h(n!) = n^{o(1)}$ as $n \to \infty$.

The main construction gives a local representation theorem in which the factorial index is close to the logarithm of the target.

Theorem 1.2. *For every $\eta > 0$ there are constants $C_{\eta}, K_{\eta} > 0$ such that every integer $R \ge 1$ is a sum of at most*

$$C_{\eta}(1 + \log\log(R+3))^{K_{\eta}}$$

distinct positive divisors of

$$\lceil C_{\eta}(\log(R+3))^{1+\eta} \rceil !.$$

There are three points to the argument. First, many short divisors can be assembled in disjoint binary blocks. This makes a distribution of short sums whose mass in every relevant residue class is small. Second, multiplicative mixing and Fourier inversion give short positive representatives of every residue

Date: September 16, 2026.

2020 *Mathematics Subject Classification.* Primary 11A25; Secondary 11B13, 11L07, 68V15.

Key words and phrases. Erdős problem, practical numbers, factorial divisors, sum-product estimates, exponential sums, formal verification, Lean.

modulo 2^k. Their height can be a fixed power of 2^k: it need not be less than the modulus. Third, choosing that modulus below a fixed power of the target makes the quotient recursion positive and gives a geometric decrease in its logarithm. Multinomial divisibility keeps the factorial cost under control throughout.

Sections 2–6 give the mathematical proof. Section 7 specifies the formal theorem, the verification environment, and the relation between this presentation and the Lean development. Appendix A explains the final analytic iteration used by the formal proof. The stronger question of a fixed polylogarithmic bound for $h(n!)$, part (c) of the problem, is not a consequence of the bounds obtained here.

Notation. All logarithms are natural unless marked $\log_2$. Write $e_Q(x) = \exp(2\pi i x/Q)$. If $d \mid Q$, let $\pi_d : \mathbb{Z}/Q\mathbb{Z} \to \mathbb{Z}/d\mathbb{Z}$ be reduction. For a probability measure ν on $\mathbb{Z}/Q\mathbb{Z}$, write $\nu(a \bmod d)$ for its mass on $\pi_d^{-1}(a)$. Separate random variables appearing in an expectation are independent unless otherwise specified. Constants depending on a fixed parameter are chosen before the target and scale vary. We use $\ll_\eta$ and $O_\eta(\cdot)$ with this convention.

2. A dyadic mixing estimate

The external analytic input to the proof presented here is the following theorem of Bourgain [1, Theorem 2].

Theorem 2.1 (Bourgain)**.** *For every $\gamma > 0$ there exist $\varepsilon_B, \tau_B > 0$ and an integer $r \geq 1$ with the following property. For all sufficiently large Q, if nonempty sets $A_1, \dots, A_r \subseteq \mathbb{Z}/Q\mathbb{Z}$ satisfy*

$$|A_i \cap \pi_d^{-1}(a)| < d^{-\gamma}|A_i| \quad (d \mid Q,\ d > Q^{\varepsilon_B},\ a \bmod d), \tag{2.1}$$

then, for every b coprime to Q,

$$\left|\mathbb{E}_{x_i \in A_i} e_Q(bx_1 \cdots x_r)\right| < Q^{-\tau_B}. \tag{2.2}$$

All constants depend only on γ.

Projection modulo a divisor need not preserve uniformity on a set. We therefore record a measure version. Only its unit-supported case will be used in the factorial construction.

Lemma 2.2 (Weighted dyadic mixing)**.** *Fix $0 < g \leq 1/2$. There are $r \geq 1$, $t > 0$, and Q_0, depending only on g, such that the following holds. Let $Q = 2^j \geq Q_0$, and let $\nu_1, \dots, \nu_r$ be probability measures on $\mathbb{Z}/Q\mathbb{Z}$ satisfying*

$$\nu_i(a \bmod 2^\ell) \leq 2^{-g\ell} \quad (2 \leq \ell \leq j,\ a \bmod 2^\ell). \tag{2.3}$$

Then for every odd b,

$$\left|\mathbb{E}_{x_i \sim \nu_i} e_Q(bx_1 \cdots x_r)\right| \leq Q^{-t}. \tag{2.4}$$

Proof. Apply Theorem 2.1 with $\gamma = g/8$, and fix the resulting ε_B, τ_B, r. Put $N = \lfloor Q^{g/2} \rfloor$. Independently for each i, sample N points with law ν_i. Let σ_i be their empirical measure, B_i the set of distinct sampled points, and $L_i = N - |B_i|$.

Every atom of ν_i has mass at most Q^{-g}. Since L_i is at most the number of coincident unordered pairs of samples,

$$\mathbb{E}L_i \leq \binom{N}{2} \sum_x \nu_i(x)^2 \leq \binom{N}{2} Q^{-g} \leq \tfrac{1}{2}.$$

Consequently Markov's inequality gives

$$\mathbb{P}\{L_i/N > Q^{-g/4}\} = O(Q^{-g/4}). \tag{2.5}$$

For $d = 2^\ell > Q^{\varepsilon_B}$, the number of samples in any fixed class modulo d is binomial with mean at most Nd^{-g}. The condition $\ell \geq 2$ holds for all these divisors when Q is large. Set $T_d = \frac{1}{4}Nd^{-g/8}$. Uniformly in $d \mid Q$,

$$T_d \gg Q^{3g/8}, \qquad \frac{Nd^{-g}}{T_d} \leq 4Q^{-7g\varepsilon_B/8} \longrightarrow 0.$$

The binomial upper-tail bound $\mathbb{P}\{X \ge T\} \le (e\mathbb{E}X/T)^T$ for $T > \mathbb{E}X$ implies that the sample count is less than T_d, except with probability $\exp(-cQ^{3g/8})$, after changing $c > 0$ if necessary. There are fewer than $2Q$ residue classes in total over all divisors of Q. A union bound makes the inequalities simultaneous for the fixed number r of measures, with exponentially small failure probability.

Combine this event with $L_i/N \le Q^{-g/4}$ for every i. Its complement has probability $O_r(Q^{-g/4})$, and on the event $|B_i| \ge N/2$ for large Q. Hence

$$|B_i \cap \pi_d^{-1}(a)| < T_d \le \tfrac{1}{2}d^{-g/8}|B_i| < d^{-g/8}|B_i|.$$

Theorem 2.1 applies to the B_i.

With total variation defined as half the ℓ^1 distance,

$$\|\sigma_i - \mathrm{Unif}(B_i)\|_{\mathrm{TV}} \le L_i/N.$$

Indeed, if $L_i > 0$, the empirical measure is the convex combination $\frac{|B_i|}{N}\,\mathrm{Unif}(B_i) + \frac{L_i}{N}\lambda_i$, where λ_i is the law of the repeated copies left after retaining one copy of each point. The assertion is immediate when $L_i = 0$. Replacing the empirical laws one at a time therefore changes an average of a complex function of modulus at most one by at most $2rQ^{-g/4}$. On the good event,

$$\left|\mathbb{E}_{x_i\sim\sigma_i} e_Q(bx_1\cdots x_r)\right| \le Q^{-\tau_B} + 2rQ^{-g/4}.$$

On its complement the modulus is at most one. Averaging over the samples gives the same bound with an error $O_r(Q^{-g/4})$. By independence across i, the expectation of the empirical product average, before taking its absolute value, is exactly the average with the original laws. Thus

$$\left|\mathbb{E}_{x_i\sim\nu_i} e_Q(bx_1\cdots x_r)\right| \le Q^{-\tau_B} + O_r(Q^{-g/4}).$$

Taking $t = \frac{1}{2}\min(\tau_B, g/4)$ and increasing Q_0 proves the lemma. □

3. Short divisors and binary blocks

It suffices to establish Theorem 1.2 for $0 < \eta \le 1$: larger η follow by increasing the factorial index. Fix such an η. For an integer scale k set

$$m = \lceil k^{1+\eta} \rceil, \qquad c = \frac{\eta}{16(1+\eta)}, \qquad D_\ell = \{0\} \cup \{d : d \mid m!,\ 1 \le d < 2^\ell\}. \tag{3.1}$$

The digit set D_ℓ will be used for $1 \le \ell \le k$.

Lemma 3.1. *For all sufficiently large k, uniformly for $1 \le \ell \le k$,*

$$|D_\ell| \ge 2^{c\ell}. \tag{3.2}$$

Proof. If $\ell \le 2\log_2 m + 2$, every integer from 1 to $\min(m, 2^\ell - 1)$ is a divisor of $m!$. Including zero gives

$$|D_\ell| \ge \min(m+1, 2^\ell) \ge 2^{\ell/3}$$

for sufficiently large m, which is enough since $c < 1/3$.

Otherwise put $a = \log_2 m$ and $v = \lfloor(\ell - 1)/a\rfloor$. The inequality $\ell > 2a + 2$ gives

$$\frac{\ell}{2a} \le v \le \frac{k}{a}.$$

Every product of v distinct primes at most m is a divisor of $m!$ and is at most $m^v \le 2^{\ell-1} < 2^\ell$. Distinct prime subsets give distinct products. Write $P = \pi(m)$. The elementary bound $P \gg m/\log m$ implies $P/v \gg m/k$. Thus, for large k, $P \ge v$ and

$$\log(P/v) \ge \tfrac{\eta}{2}\log k$$

uniformly in the indicated range of ℓ. Since $\binom{P}{v} \geq (P/v)^v$ and $\log m \leq 2(1+\eta)\log k$, we get

$$\begin{aligned}\log|D_\ell| &\geq v\log(P/v)\\ &\geq \frac{\ell\log 2}{2\log m}\,\frac{\eta}{2}\log k \geq \frac{\eta}{8(1+\eta)}\ell\log 2 \geq c\ell\log 2.\end{aligned}$$

The counted products are squarefree, so there is no further multiplicity condition to check for divisibility by $m!$.

For completeness, the prime-counting bound follows from

$$\frac{2^m}{m+1} \leq \binom{m}{\lfloor m/2\rfloor} \leq m^{\pi(m)}.$$

For the second inequality, the contribution of each power of a prime to the valuation of the binomial coefficient is zero or one. A given prime therefore contributes at most a factor m. □

Set $q = 2^k$ and $b_0 = \lfloor \log_2 m \rfloor$. For sufficiently large k we have $2 \leq b_0 < k$. Choose an initial digit uniformly from the odd integers in $[1, 2^{b_0})$. Starting with the endpoint $b = b_0$, append a block of length

$$\ell = \min(b, k-b)$$

by independently choosing a uniform digit from D_ℓ, multiplied by 2^b. Replace b by $b+\ell$ and continue until $b = k$. Let X be the sum and let A_k be the set of its possible values.

The binary positions of the blocks are disjoint and every digit is smaller than its radix. The digit choices therefore determine distinct integers, so X is uniform on A_k, and

$$A_k \subseteq [1, 2^k) \cap (2\mathbb{Z}+1). \tag{3.3}$$

Write $J(k)$ for the number of blocks, including the initial one. Since the endpoint doubles except possibly at the last step, $J(k) = O(\log k)$.

Every initial digit divides $m!$. Each subsequent nonzero digit contributes $2^b d$ with $d \mid m!$ and $b \leq k$. The $2k$ consecutive integers after m include at least k even integers. Consequently

$$2^k m! \mid (m+2k)!. \tag{3.4}$$

With $M = m + 2k$, each $X \in A_k$ is thus a sum of at most $J(k)$ positive divisors of $M!$. A zero digit contributes no summand. Repetitions among these divisors are permitted at this stage.

Lemma 3.2 (Residue bounds)**.** *Let $g = c/4$. For all sufficiently large k, the law ν_k of X satisfies*

$$\nu_k(a \bmod 2^j) \leq 2^{-gj} \quad (2 \leq j \leq k,\ a \bmod 2^j). \tag{3.5}$$

For $j \leq b_0$, its projection is exactly uniform on the odd residues modulo 2^j.

Proof. All shifts after the initial block are multiples of 2^{b_0}. The last assertion follows, and the maximal mass for $2 \leq j \leq b_0$ is $2^{1-j} \leq 2^{-gj}$ because $g \leq 1/2$.

Now let $j > b_0$, and let $b \leq j$ be the last completed block endpoint at or below j. If $b < j$, the next block has length at most b; hence always $b \geq j/2$. Specifying a residue modulo 2^j fixes every completed block digit. By Lemma 3.1, its probability is at most

$$2^{-(b_0-1)} \prod_{\text{completed later blocks}} |D_\ell|^{-1} \leq 2^{1-cb} \leq 2^{1-cj/2}.$$

For large k, we have $b_0 \geq 4/c$. Since $j > b_0$, the last expression is at most $2^{-cj/4} = 2^{-gj}$. □

The residue estimate is a property of the constructed short sums. It requires no distribution theorem for smooth numbers in arithmetic progressions.

4. Short positive representatives of every residue

Keep η, c, g fixed and apply Lemma 2.2 to obtain r, t, Q_0. These constants are fixed before k varies. Let $X_1, \dots, X_r$ be independent with law ν_k, and let Z be independent and uniform on $\{0, 1\}$. Define

$$Y = X_1 \cdots X_r + Z,$$

and let μ_k be its residue law modulo $q = 2^k$. We use the Fourier normalization $\widehat{\mu}_k(a) = \mathbb{E}e_q(aY)$, so that $\widehat{\mu}_k(0) = 1$.

Every nonzero frequency a mod 2^k has a unique conductor 2^j, where

$$a = 2^{k-j}a', \qquad a' \text{ odd}, \qquad 1 \le j \le k.$$

There are 2^{j-1} frequencies of conductor 2^j. For $j = 1$, the independent digit Z makes the Fourier coefficient zero. For $2 \le j \le b_0$, the projected X_i are uniform on the group of odd residues modulo 2^j, so their product is uniform on the same group. At an odd frequency,

$$\sum_{\substack{x \bmod 2^j \\ x \text{ odd}}} e_{2^j}(a'x) = 0.$$

Thus these Fourier coefficients also vanish.

For $j > b_0$, the projected laws satisfy (2.3) by Lemma 3.2. Since $2^{b_0+1} \to \infty$, the threshold Q_0 is met simultaneously for all such j once k is large. The factor contributed by Z has modulus at most one. Lemma 2.2 therefore gives

$$|\widehat{\mu}_k(a)| \le 2^{-tj} \qquad (j > b_0). \tag{4.1}$$

Choose a fixed integer $s \ge 1$ with $st \ge 2$. The law of a sum of s independent copies of Y is μ_k^{*s}, and

$$\sum_{a \ne 0} |\widehat{\mu}_k(a)|^s \le \sum_{j=b_0+1}^{k} 2^{j-1}2^{-stj} < 1. \tag{4.2}$$

Fourier inversion, with the real part of the nonconstant contribution bounded below by minus its absolute value, yields

$$\mu_k^{*s}(a) \ge \frac{1}{q}\left(1 - \sum_{b \ne 0} |\widehat{\mu}_k(b)|^s\right) > 0$$

for every residue a mod q.

To retain actual integer witnesses, expand each product $X_1 \cdots X_r$. It is a sum of at most $J(k)^r$ positive terms, each dividing $(M!)^r$. The multinomial coefficient $(rM)!/(M!)^r$ is an integer. Thus every term divides $(rM)!$. Put

$$H(k) = r(\lceil k^{1+\eta} \rceil + 2k), \qquad D = r + 1. \tag{4.3}$$

Proposition 4.1 (Modular representatives)**.** *For all sufficiently large k, each residue a mod 2^k has an integer representative S such that*

$$S \equiv a \pmod{2^k}, \qquad 0 \le S \le (2^k)^D, \tag{4.4}$$

$$S \text{ is a sum of at most } s(J(k)^r + 1) \text{ positive divisors of } H(k)!, \tag{4.5}$$

$$2^k \mid H(k)!. \tag{4.6}$$

In particular the summand count is $O_\eta((\log k)^r)$ and the factorial index is $O_\eta(k^{1+\eta})$.

Proof. Positivity of $\mu_k^{*s}(a)$ supplies a tuple of the original integer values whose sum S has the desired residue. Its product expansions use at most $s(J(k)^r + 1)$ summands, including the optional ones from Z. The preceding divisibility argument puts them all in $H(k)!$. Moreover,

$$S \le s(q^r + 1) \le q^{r+1}$$

once $q \ge 2s$. Finally, (3.4) gives $2^k \mid M! \mid H(k)!$. □

5. Positive lifting and distinct summands

Lemma 5.1 (Removing repetitions)**.** *If an integer $R > 0$ is a sum of $u \geq 1$ positive divisors of $F!$, with repetitions allowed, then it is a sum of at most u distinct positive divisors of $(F + 2u)!$.*

Proof. Repeatedly replace a pair of equal summands d, d by the single summand $2d$. Each replacement preserves the sum and reduces the number of terms, so the process stops after at most $u-1$ replacements. Tracing any final term backwards shows that it is 2^a times an original divisor, for some $a \leq u - 1$. All final terms therefore divide $2^{u-1}F!$. Since the $2u$ consecutive integers after F include at least u even integers, $2^{u-1}F! \mid (F + 2u)!$. At termination the terms are distinct. □

Proof of Theorem 1.2. Fix a threshold large enough for Proposition 4.1 and the inequalities below. For a target R above this threshold, write

$$L = \log_2 R, \qquad k = \left\lfloor \frac{L}{D+1} \right\rfloor, \qquad q = 2^k.$$

Choose the representative S of R mod q given by the proposition. Its height satisfies

$$S \leq q^D \leq R^{D/(D+1)} \leq R/2.$$

Consequently

$$R = S + qR', \qquad R' = (R - S)/q \in \mathbb{Z}_{>0}. \tag{5.1}$$

The modulus is at least $R^{1/(D+1)}/2$, so

$$\log_2 R' \leq \frac{D}{D+1} \log_2 R + 1.$$

When $L \geq 2(D + 1)$, this is at most ρL, where

$$\rho = 1 - \frac{1}{2(D+1)} < 1.$$

Repeat (5.1) until the target is below the fixed threshold. Let the successive targets be $R_0 = R, R_1, \ldots, R_v$, the moduli $q_i = 2^{k_i}$, and the representatives S_i. The logarithms decrease geometrically. It follows that

$$v = O_\eta(1 + \log\log(R + 3)), \tag{5.2}$$

$$\sum_{i<v} H(k_i) = O_\eta((\log(R + 3))^{1+\eta}). \tag{5.3}$$

Indeed, $H(k_i) \ll_\eta (\log R_i)^{1+\eta}$, and these upper bounds form a geometric series. The total number u of expanded terms, including the terminal remainder, is at most

$$u = O_\eta((1 + \log\log(R + 3))^{r+1}), \tag{5.4}$$

by (5.2) and the summand bound in Proposition 4.1.

The scaling factors in the unrolled recursion also have to be accounted for in the factorial. Put $P_j = \prod_{i<j} q_i$, with $P_0 = 1$. Then

$$R = \sum_{j<v} P_j S_j + P_v R_v. \tag{5.5}$$

Choose a fixed positive integer T at least as large as every possible terminal remainder. A term in S_j divides $H(k_j)!$, while every preceding q_i divides its own factorial $H(k_i)!$. Thus each expanded term in (5.5) divides

$$T! \prod_{i<v} H(k_i)!.$$

The terminal term satisfies the same assertion because $R_v \mid T!$. Multinomial integrality shows that this product divides $F!$, where

$$F = T + \sum_{i<v} H(k_i).$$

In particular all terms divide one factorial whose index is $O_\eta((\log(R+3))^{1+\eta})$.

Apply Lemma 5.1 to the u terms. Their sum becomes a sum of at most u distinct divisors of $(F+2u)!$. Equations (5.3) and (5.4) imply

$$F + 2u = O_\eta((\log(R+3))^{1+\eta}).$$

Increasing the constant handles the finitely many smaller targets, each represented by itself in a sufficiently large fixed factorial. We may take $K_\eta = r + 1$ and increase C_η to cover both bounds in the statement. □

The choice of the modulus in (5.1) is what makes the height allowance in Proposition 4.1 usable: the representative may exceed q, but it is smaller than the current target. At no stage is a negative summand introduced.

6. The factorial radix recurrence

First, every factorial is practical. To see this, induct on n, starting with $0! = 1$. For $0 \le R \le n!$, write $R = na + r$, with $0 \le r < n$. By induction, $a \le (n-1)!$ has a representation by distinct divisors of $(n-1)!$, with the empty sum used if $a = 0$. Multiplying by n gives distinct divisors of $n!$, all at least n. If $r > 0$, it is itself a divisor of $n!$ and is smaller than n, so it may be added as a new summand. This also proves that (1.1) is finite at every factorial.

Fix $0 < \varepsilon < 1$, and choose $0 < \eta \le 1$ so that

$$(1-\varepsilon)(1+\eta) < 1; \tag{6.1}$$

for example, $\eta = \varepsilon/2$ suffices. Theorem 1.2 implies that, for all sufficiently large n, every integer R with

$$0 \le R \le \exp(n^{1-\varepsilon}) \tag{6.2}$$

is a sum of $O_\varepsilon((\log n)^{K_\varepsilon})$ distinct divisors of $n!$, uniformly in R. For $R > 0$, the local factorial index is at most

$$O_\varepsilon(n^{(1-\varepsilon)(1+\eta)}) = o(n),$$

and the local summand bound has the stated size. For $R = 0$ use the empty sum.

For large n, put

$$b(n) = \left\lfloor \frac{n^{1-\varepsilon}}{4\log n} \right\rfloor, \qquad Q_n = \frac{n!}{(n-b(n))!}. \tag{6.3}$$

We have $1 \le b(n) < n$ and

$$\log Q_n \le b(n)\log n \le n^{1-\varepsilon}/4.$$

Given $0 \le R \le n!$, write $R = Q_n a + r$ with $0 \le r < Q_n$. Then $a \le (n-b(n))!$. Its shortest representation by divisors of that factorial, multiplied by Q_n, uses distinct divisors of $n!$, all at least Q_n. Represent r by the uniform local result (6.2). Every positive term in this representation is less than Q_n because the terms sum to $r < Q_n$. The two collections are therefore disjoint. Taking the maximum over R gives

$$h(n!) \le h((n-b(n))!) + O_\varepsilon((\log n)^{K_\varepsilon}). \tag{6.4}$$

Proof of Theorem 1.1. Iterate (6.4) until the current index is below its fixed threshold. While that index is in $(X/2, X]$, its decrement is at least $c_\varepsilon X^{1-\varepsilon}/\log X$, for sufficiently large X. Thus there are $O_\varepsilon(X^\varepsilon \log X)$ steps in this band, each costing $O_\varepsilon((\log X)^{K_\varepsilon})$. Summing over the bands $X = n, n/2, n/4, \ldots$, and absorbing the bounded terminal value, yields

$$h(n!) \ll_\varepsilon n^\varepsilon (\log n)^{K_\varepsilon+1}. \tag{6.5}$$

For a prescribed $\delta > 0$, choose $0 < \varepsilon < \min(1, \delta)$. The right side of (6.5) is $o(n^{\delta})$. Hence $h(n!) < n^{\delta}$ for every sufficiently large n. □

Remark 6.1 (Order of parameters). The constants are chosen in the order

$$\delta \longrightarrow \varepsilon \longrightarrow \eta \longrightarrow c, g \longrightarrow r, t, Q_0 \longrightarrow s, D \longrightarrow \text{thresholds}.$$

The scale k, the local target R, and the final index n vary only after these choices. No uniform bound in η is asserted. In particular, the dependence of K_{ε} and the constants on ε in (6.5) does not give $h(n!) \ll (\log n)^{C}$ for a fixed C.

7. Lean formalization and reproducibility

The accompanying development uses Lean 4 [6] and Mathlib [7]. Its target is the declaration `Erdos18.erdos_18b` in the pinned Formal Conjectures library [5]. The formal definition of `Erdos18.practicalH` takes the minimum cardinality of a divisor set containing a subset with the required sum. This agrees with (1.1): the representing subset itself is a permissible divisor set, and its cardinality is no larger than that of the containing set. Factorial practicality supplies a representation for every target.

In the challenge's specified answer mode, the exact type is

$$\mathsf{True} \longleftrightarrow \forall \varepsilon \in \mathbb{R}, \quad \varepsilon > 0 \implies \forall^{\text{eventually}} n \in \mathbb{N}, \quad h(n!) < n^{\varepsilon},$$

where "eventually" is with respect to the filter at infinity on $\mathbb{N}$. The submitted theorem closes with the following code:

```
theorem target : fcTypeOfName% "Erdos18.erdos_18b" := by
  exact erdos18b_of_weighted_dyadic_mixing weighted_dyadic_mixing
```

The theorem `weighted_dyadic_mixing` proves its mixing premise; it is not an axiom or an assumption left to the user.

7.1. The analytic and arithmetic components.

The written proof in Section 2 obtains the measure estimate from Bourgain's published theorem. The formal development proves the needed odd-law specialization internally. It establishes hereditary sum-product growth modulo powers of two, inverse additive-energy bounds, control of additive stabilizers, and quantitative flattening. Weighted unit averaging and a Fourier iteration then give the desired mixing statement. The analytic organization is informed by sum-product and flattening methods, including those in the appendix to Bourgain–Gamburd [2].

The three final analytic steps are recorded in the declarations

```
exists_dyadic_weighted_probability_flattening,
exists_dyadic_probability_flattening_iteration,
exists_dyadic_unit_product_fourier_decay.
```

Appendix A describes their mathematical relationship. The resulting mixing theorem specializes this estimate to projections of finite sets of odd natural numbers. It retains the multiplicities created by reduction modulo 2^{j}.

The arithmetic reduction is `erdos18b_of_weighted_dyadic_mixing`. Its proof includes the divisor digits, binary blocks, conductor transfer, positive modular representatives, quotient recursion, duplicate merging, and factorial radix argument. The formal asymptotic interfaces often replace explicit powers of logarithms by arbitrarily small positive powers, which suffice for the exact target. The constants and the sharper local formulation in Theorem 1.2 are presented mathematically above; the audited final Lean statement is precisely Theorem 1.1.

7.2. **Files and checks.** All mathematical declarations are contained in a single submission body, `Main.lean`. The challenge supplies its imports and outer `Bounty` namespace. A standalone copy, `Challenge.lean`, contains the same body together with

```
import FormalConjectures.ErdosProblems.«18»
import TaskSupport

namespace Bounty
```

and the closing `end Bounty`. Thus the two files are alternative packagings of the same proof, and neither imports another local mathematical proof file.

The checked environment is:

> Lean 4.33.1, revision
> `819816b2e0a3bf405af45ae5c7af2491d8f5bee6`;
>
> Formal Conjectures, revision
> `8432eac998110a563e03df65a28c117e97c8c142`;
>
> Mathlib, revision
> `0df444a360eaa60ab8c11dca51a86af692955474`.

The source conjecture is compiled in the required `google.answer=always_true` mode; its pinned source is unchanged. The submission uses default compiler limits and contains no compiler option overrides.

The file contains 806 theorem declarations, including `Bounty.target`. The completed local checks comprise compilation of the exact trusted wrapper, a separate check of the explicit target type, a transitive axiom audit of every theorem, and a dependency traversal of the final target. The latter visits 25,688 declarations and finds no dependency on the admitted source theorem `Erdos18.erdos_18b`. It also confirms that the optional, stronger `BourgainDyadicSetEstimate` is not used in the final proof.

There are no admitted steps or added axioms. The target's only transitive axioms are Lean's standard `propext`, `Classical.choice`, and `Quot.sound`. The original challenge static scanner accepts the 735,691-byte submission body. Its SHA-256 digest is

> `4d6ee89617415288e9f127beb3c0e0211c011f891889fcb43d0e16373bd14088`.

The accompanying verification report records the compiler exits, source hashes, exact-type check, and audits. These checks were performed locally; the hosted challenge Comparator has not been run. Formal verification of the specified statement and independent review of this expository manuscript are distinct checks.

Appendix A. From weighted flattening to unit mixing

This appendix describes the internal analytic bridge used in the Lean development. It also explains why no reduction to general nonunit inputs is needed for the construction in Section 3. We write $q = 2^J$, work on the additive group $\mathbb{Z}/q\mathbb{Z}$, and set

$$E(f) = \sum_x |f(x)|^2, \qquad \widehat{f}(k) = \sum_x f(x) e_q(kx), \qquad D_u f(x) = f(u^{-1}x)$$

for a unit u. Thus Parseval reads $\sum_k |\widehat{f}(k)|^2 = qE(f)$.

For fixed $\gamma, \epsilon > 0$, say that a law has the residue bound (γ, ϵ) if its mass on every class modulo 2^L is at most $2^{-\gamma L}$ whenever $\epsilon J < L \le J$. The same definition applies to a law on the units, using reduction of its unit values. Translation, reflection, unit dilation, probability mixtures, and convolution with a probability law preserve the relevant upper bounds. For convolution this follows by conditioning on one input; for a mixture it follows by averaging the corresponding fiber inequalities.

A.1. **The checked flattening estimate.** The formal development proves the following statement. For every $\gamma > 0$ and $0 < \delta \le 1$ there are $0 < \epsilon \le 1/4$, $\tau > 0$, and a threshold such that, for all sufficiently large J, probability laws p and w on $\mathbb{Z}/q\mathbb{Z}$ and its units, respectively, satisfying the residue bound (γ, ϵ) obey

$$E(p) \ge q^{-(1-\delta)} \quad \Longrightarrow \quad \sum_u w(u) E(p * D_u p) \le q^{-\tau} E(p). \tag{A.1}$$

The full formal theorem also allows subprobability laws. Its proof passes from hereditary sum-product growth to an obstruction for additive stabilizers, combines this with inverse energy, and decomposes both nonuniform laws into dyadic levels. Explicit bounds control the discarded levels and the small-weight tails. All thresholds are chosen before J, p, w.

A.2. **An additive iteration.** Let f be an initial probability and $w_0, w_1, \dots$ unit probability laws, all satisfying the same residue bound (γ, ϵ) from (A.1). Put $\widetilde{f}(x) = f(-x)$ and define

$$\nu_0 = f * \widetilde{f}, \qquad \theta_i = \sum_u w_i(u) D_u \nu_i, \qquad \nu_{i+1} = \theta_i * \theta_i. \tag{A.2}$$

Define in parallel the multiplicative product laws

$$P_0 = f, \qquad P_{i+1} = \sum_u w_i(u) D_u P_i.$$

All laws in (A.2) remain probability laws and retain the input residue bound. Their Fourier transforms are real and nonnegative: this starts with $\widehat{\nu}_0 = |\widehat{f}|^2$, is preserved by unit mixtures, and is preserved by squaring.

Convexity of the squared ℓ^2 norm gives

$$\begin{aligned} E(\nu_{i+1}) &\le \sum_{u,v} w_i(u) w_i(v) E(D_u \nu_i * D_v \nu_i) \\ &= \sum_{u,v} w_i(u) w_i(v) E(\nu_i * D_{u^{-1}v} \nu_i). \end{aligned} \tag{A.3}$$

For each fixed u, the relative law $z \mapsto w_i(uz)$ has the same residue bound. If $E(\nu_i) \ge q^{-(1-\delta)}$, apply (A.1) to the inner average in (A.3) to get $E(\nu_{i+1}) \le q^{-\tau} E(\nu_i)$. At every energy level, convolution with a probability law and convexity give $E(\nu_{i+1}) \le E(\nu_i)$. As $E(\nu_0) \le 1$, a fixed integer R with $R\tau \ge 1$ therefore ensures

$$E(\nu_R) \le q^{-(1-\delta)}. \tag{A.4}$$

Once the threshold is crossed the energy cannot increase above it; if it is not crossed earlier, the multiplicative decrease proves the bound at step R.

A.3. **Fourier domination and the last multiplier.** The pointwise comparison is

$$|\widehat{P}_i(k)|^{2^{i+1}} \le \widehat{\nu}_i(k). \tag{A.5}$$

For $i = 0$ it is equality. Given the assertion at i, the triangle inequality and Jensen's inequality for the power 2^{i+1} give

$$|\widehat{P}_{i+1}(k)|^{2^{i+1}} \le \sum_u w_i(u) |\widehat{P}_i(uk)|^{2^{i+1}} \le \widehat{\theta}_i(k).$$

Squaring proves the induction step, since $\widehat{\nu}_{i+1} = \widehat{\theta}_i^2$.

Take one additional unit law w_R. Its residue bound at $L = J$ gives $w_R(u) \le q^{-\gamma}$. For a unit frequency k, Cauchy–Schwarz and the injectivity of $u \mapsto uk$ on the units give

$$\begin{aligned}|\widehat{\theta}_R(k)|^2 &\le \sum_u w_R(u)|\widehat{\nu}_R(uk)|^2 \\ &\le q^{-\gamma}\sum_\xi |\widehat{\nu}_R(\xi)|^2 = q^{1-\gamma}E(\nu_R) \le q^{\delta-\gamma}.\end{aligned}$$

The unsquared comparison in the preceding induction then yields

$$|\widehat{P}_{R+1}(k)|^{2^{R+2}} \le q^{\delta-\gamma}.$$

Choosing $\delta = \min(\gamma/2, 1/2)$ gives positive cancellation exponent

$$t = \frac{\gamma-\delta}{2^{R+2}} > 0.$$

There are $R + 2$ inputs, counting the initial law. This is the analytic conclusion used to prove `weighted_dyadic_mixing`: specialize all inputs to the same odd law obtained by reducing a finite set of natural numbers. The initial nonconcentration bounds at every level $L \ge 2$ imply the cutoff bounds for large J. The support in the units follows from oddness. This establishes the premise required by the arithmetic reduction without a general nonunit extension.

Acknowledgments and AI assistance

The authors acknowledge substantial assistance from OpenAI's Codex in mathematical exploration, development and debugging of the Lean proofs, computational and dependency checks, and preparation of this manuscript. The local Lean checks described in Section 7 provide the verification evidence for the formal statement. The authors also thank the developers and contributors of Lean, Mathlib, and Formal Conjectures for the underlying libraries and tools.

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